BAIT 508 GROUP PROJECT

B A 2 | G R O U P 27

TEAM MEMBERS - STUDENT ID

Role description

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  Thao studied the data and implemented the analysis coding across all project components and worked on the report for Part 1 and 3

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  Koko studied the data and implemented the analysis coding for Part 1 and 2, and worked on the report for Part 2 and 3

PROJECT REPORT

This report was created with the following steps:

1. **Preliminary data processing**

At the start of the project, we were given 3 data sets:

- "**major_groups.csv**": a list of major SEC-categorized sectors and their corresponding codes,
- "**public_firms.csv**": historical data of IPO companies within those sectors,
- "**2020_10K_item1_full.csv**": a sample of 5,988 firms and their Business Description (item 1) text in their 2020 10-K reports.

As our sector of interest is Transportation By Air (group code: 45), we then filtered the data in **public_firms.csv** to select just the firms from the above sector. There is a Python method called *startswith(),* which can return True or False values when being applied on a string. Specifically, it will tell users if the string begins with a specified set of characters or not. However, we are working with a data frame here and we cannot directly call such a string method on the entire column of a Panda Series. To work around this, we defined a user-defined function that applies startswith("45") to each row value of the column containing the SIC code in the data table. Then, pandas evaluates row by row and returns only companies whose SIC code starts with "45".

```
import pandas as pd

df = pd.read_csv('public_firms.csv', index_col=None)

def starts_with_45(x):

   return str(x).startswith("45")

company_data = df[df["sic"].apply(starts_with_45)]
```

For the next step, we conducted an exploratory data analysis of the relevant data we have and found:

- **27 unique firm-year observations** in the data set, indicating the longest continuous 10-K reporting period is 27 years.
- **111 unique firms in the filtered data set,** which is reasonable as we are focusing on a relatively niche industry versus broader ones e.g., retail.

- **Only 4 firms with full records over 27 years (1994-2000). Potential reasons might be:** 1) the majority of the firms went public after 1994, 2) firms exited the industry before 2020, or 3) Missing reports or other data issues, which may apply to international businesses.

2. **Preliminary quantitative analysis**

This dynamic and concentrated industry overview prompted us to do a more targeted analysis to understand the industry's top performers and how industry shocks affected their performance, given Transportation by Air is a cyclical sector.

From the Transportation by Air dataset, we filtered to get only the 2020 data records and used the *sort_values()* function to sort their stock prices by descending order.

The top 10 firms with the highest stock price in 2020 result is split quite evenly among cargo service providers, passenger airlines and airport operators. FedEx was traded at the highest with $259.62 per stock, which was thanks to the E-commerce and global shipping boom during the COVID pandemic. For airlines, while Southwest and Atlas serve primarily the U.S. domestic routes, their revenue growth and thus valuations cannot compete with international players such as Ryanair and Atlas Air with an international network.

```
top 10 firms with the highest stock price in 2020 are:
14369                    FEDEX CORP  259.62
190505           ALLEGIANT TRAVEL CO  189.24
169375   GRUPO AEROPORTUARIO SURESTE  164.93
188222  GRUPO AEROPORTUARIO DEL PACI  111.29
138511          RYANAIR HOLDINGS PLC  109.98
187501              COPA HOLDINGS SA   77.23
119487  ATLAS AIR WORLDWIDE HLDG INC   54.54
900             ALASKA AIR GROUP INC   52.00
191794  GRUPO AEROPORTUARIO DEL CENT   51.68
36248             SOUTHWEST AIRLINES   46.61
```

However, the 2020 snapshot may give us a limited view on the companies consistently outperforming in stock prices as there were some inherent post-COVID effects during that time. Therefore, we looked into the data again, row by row, and identified the top 10 companies with the highest stock price historically. Interestingly, FedEx stood out to be a long-term leader, holding the top spots for several consecutive years while some short-term winners like Ryanair, Grupo from top 10 of 2020 did not even appear in the historical ranking.

```
Top 10 firms with the highest sales in the entire history of the data set are:
14369                   FEDEX CORP  83959.0   2020
14367                   FEDEX CORP  69693.0   2018
14368                   FEDEX CORP  69217.0   2019
14366                   FEDEX CORP  65450.0   2017
14365                   FEDEX CORP  60319.0   2016
14364                   FEDEX CORP  50365.0   2015
14363                   FEDEX CORP  47453.0   2014
11301          DELTA AIR LINES INC  47007.0   2019
142    AMERICAN AIRLINES GROUP INC  45768.0   2019
1441         AMERICAN AIRLINES INC  45761.0   2019
```

Geographical distribution

When studying the sector's historical stock price leaders, we observed that they are all American companies. Such concentration at a country level left us curious about the location distribution of the Transportation by Air companies. Since the database is collected from firms filing for the SEC 10-K, we hypothesize that the majority of them are U.S-based. To explore this further, we next grouped the company data by locations and firm names, and used groupby(level=0) on the hierarchical index as well as the count() function to determine the number of unique companies per country. This approach allowed us to avoid double-counting as our data set contains repeated company rows (each row is one year observation of a company).

```
loc_distribution = company_data.groupby("location")["conm"].value_counts()
unique_loc = loc_distribution.groupby(level=0).count().sort_values(ascending=False)
```

On a high level, diversity is highly presented with the 16 countries where the firms are headquartered in. In reality, most of the companies are from North America while those from Latin America, Europe, and Asia have rather limited representations. Of which, the U.S is the most popular location, accounting for approximately 78% of all Transportation by Air companies. But if even we look into the top 10 most popular geographies, the

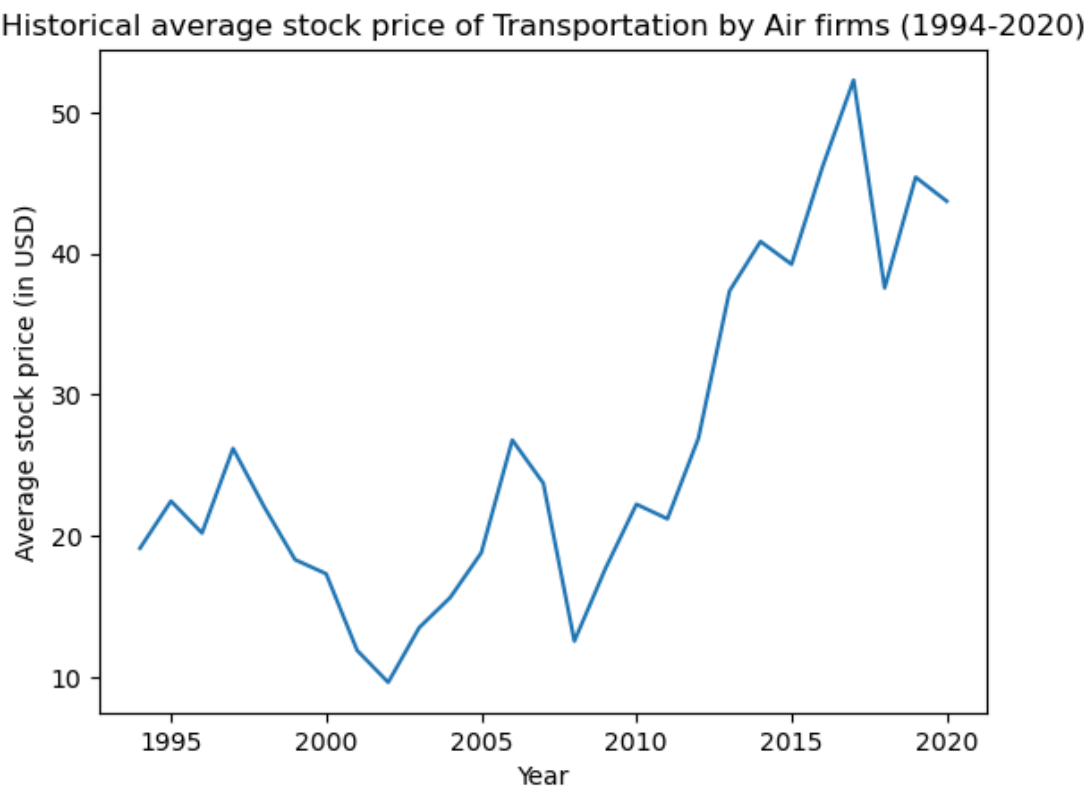
distribution still remains highly skewed to North America, with the USA dominating followed by Mexico.

```
print("Top 10 locations with the highest number of firms are: ")
print(unique_loc.head(10))
```

```
Top 10 locations with the highest number of firms are:
location
USA    87
MEX     4
BRA     3
CHN     2
COL     1
CYM     1
DEU     1
ESP     1
FRA     1
CAN     1
Name: count, dtype: int64
```

Upon the high concentration of Transportation by Air companies in North America, especially the U.S, we hypothesize that these companies are exposed to similar regulatory environments and market conditions. We thus became motivated to understand how the industry stock performance evolved over time. For this perspective, we then grouped the database by year and calculated the annual average stock price. We visualized the sector's annual average stock price during 1994-2020 in a line chart.

```
import matplotlib.pyplot as plt

ave_stock = company_data.groupby("fyear")["prcc_c"].mean()
ave_stock.plot(y=['prcc_c'], x=['fyear'], kind="line")
plt.ylabel('Average stock price (in USD)')
plt.xlabel('Year')

plt.title('Historical average stock price of Transportation by Air firms (1994-2020)')
plt.show()
```



Overall, Transportation by Air is a **volatile** industry, evidenced by the ups and downs across the 26-year lifespan. In other words, this sector is more sensitive to economic recessions and unsystematic risks. Specifically, we identified 4 key periods showcasing the industry specific dynamics from the chart:

a. **Post-9/11 aviation downturn (2001-2002):** The 9/11 attack caused a significant drop in air travel, forcing airlines to structurally revamp their strategies, from simplifying operations to expanding low-cost models to address post-trauma consumer behavior (Notis, 2012).

b. **Global financial crisis (2007-2008)**: This event, compounded with soaring fuel prices, deteriorated demand for air travel and companies' creditworthiness, causing widespread bankruptcies (Dobbs, 2009). To examine the scale of the crisis, we calculated the percentage drop in stock price between 2007 and 2008 of all relevant companies. We filtered for companies that traded in both years only, extracted their stock values, and calculated a year-over-year percent change using a *for* loop. The results were stored in an empty list and later converted to a data frame for sorting and highlighting the largest declines.

```python
#filter for companies with stock price in 2007 and 2008
stock_2007 = company_data[company_data["fyear"] == 2007][["conm", "prcc_c"]]
stock_2008 = company_data[company_data["fyear"] == 2008][["conm", "prcc_c"]]

#create an empty list to store the results
result = []

for x in stock_2007["conm"]:
    if x in stock_2008["conm"].values:
        price_08 = stock_2008[stock_2008["conm"] == x]["prcc_c"].values[0]
        price_07 = stock_2007[stock_2007["conm"] == x]["prcc_c"].values[0]
        percent_drop =  round((price_08 - price_07)/ price_07 * 100, 2)
        result.append((x, percent_drop))

df_result = pd.DataFrame(result, columns=["company", "percent drop"])
df_sorted = df_result.sort_values("percent drop", ascending=True)
print(df_sorted.head(10))            #to get top 10 most affected
```

The magnitude of the 2008 financial crisis is evident in the most affected companies' stock prices, many of which decreased 80-90% versus their 2007 performance and almost approached the penny-stock levels. These most hitters heavily concentrate in passenger airline businesses, primarily regional or smaller carriers such as Frontier, Mesa, and Pinnacle. Specifically, their small operation scales and limited financing capacity made them much vulnerable to market shocks. Some international carriers such as China Southern Airlines, China Eastern Airlines Corp and Gol Linhas may have also been impacted by currency fluctuations or weaker international flight demand. Only 2 non-airline players made it to the top 10 namely Air Transport Services Group and Macquarie Infrastructure Holding LLC, which is likely due to their focus in cargo or other aviation services.

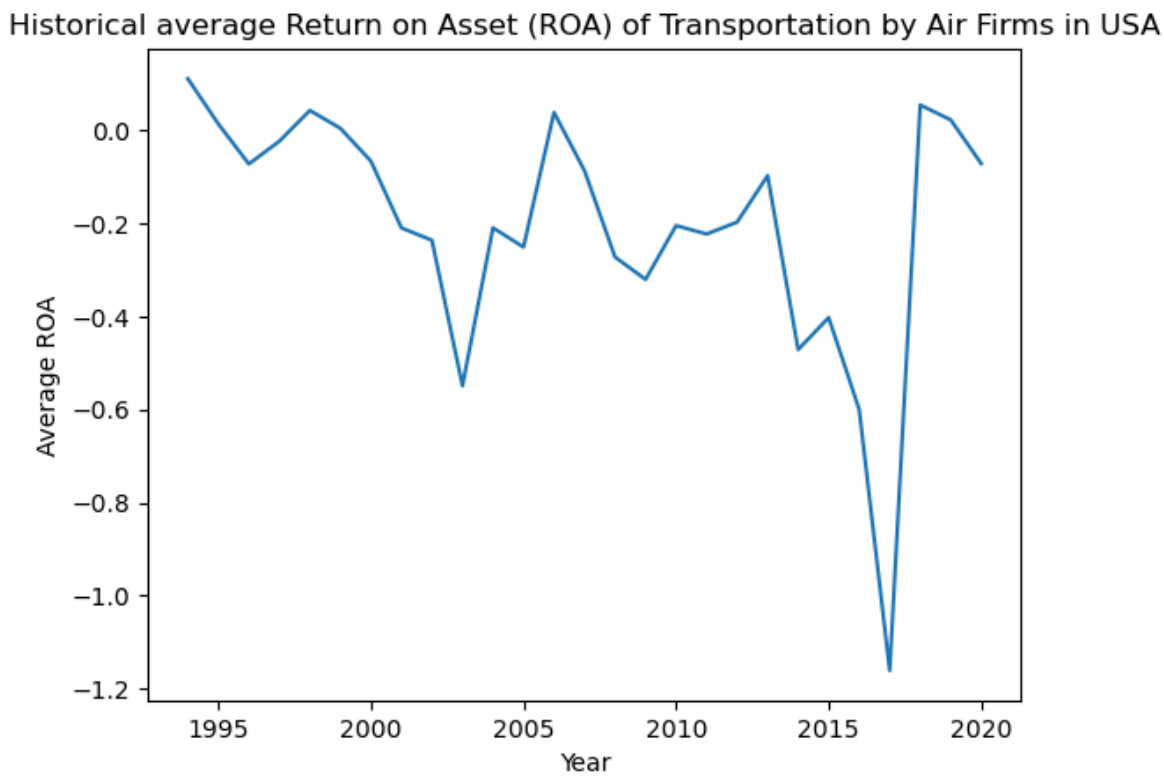
company	percent drop
FRONTIER AIRLINES HOLDINGS	-96.58
AIR TRANSPORT SERVICES GROUP	-95.69
MESA AIR GROUP INC	-91.67
MACQUARIE INFRASTR HOLD LLC	-90.70
PINNACLE AIRLINES CORP	-88.85
CHINA SOUTHERN AIRLINES	-87.03
AVANTAIR INC	-85.85
CHINA EASTERN AIRLINES CORP	-84.33
SAKER AVIATION SERVICES INC	-84.21
GOL LINHAS AEREAS INTELIGENT	-82.96

Frontier Airlines Holdings was the firm affected the most by the 2008 Financial Crisis, with an almost 97% decline in stock price between 2007 and 2008.

c. **Bull period (2009-2016):** Optimistically, this is the longest bull phase for Transportation by Air sector within the observed period. Overall companies recovered steadily from the 2008 crisis with increasing flight demand, fuel price stabilization and industry consolidations such as the mergers of United Airlines and Continental Airlines (2010), American Airlines and US Airways (2013). Altogether, these factors contributed to a sustained recovery in stock price, with an uplift of approximating 60%).

d. **COVID Pandemic (2019-2020)**: The COVID-19 outbreak led to an unprecedented collapse in travel, causing over $300-billion in lost airline annual revenue for 2 consecutive years (ICAO, n.d.). Unlike earlier crises, its slower recovery is reflected in deeper subsequent stock decline.

Lastly, we analyzed the firms' profitability via the metric Return-on-assets (ROA), which measures how much profit a company can generate per every 1 dollar-worth of assets. Since the U.S firms make up the majority of our data set and the goal of the project is to understand sector trends in the U.S market, we restricted the average historical ROA analysis to just U.S. firms. Following a similar approach above, we filtered for U.S.-only data, grouped it by year, and took the mean out of all ROA values.



Again, we observed the structural volatility of the Transportation by Air industry shown by consistently low and much often negative ROAs. Interestingly, even in the phase of recovery, its ROA still hovered close to 0, suggesting a thin margin compared to other sectors. This actually reflects the capital-intensive nature of the industry that required high fixed costs, which, when complemented with cyclical demand and sensitivity to market shocks, significantly impacted the bottom-line. Next, the market shock impacts are also visible in the steep ROA drops across the three aforementioned crisis periods. Post-crisis recoveries seemed not to last very long; for example, following the 9/11 and 2008 crisis, ROA only increased modestly for 2-5 years before taking another dip.

3. Text cleaning and NLP analysis

a. Text cleaning

We followed the following steps to clean and transform the item 1 description from the 10-K reports of the Transportation by Air sector:

- Load the data.

```
dff = pd.read_csv('2020_10K_item1_full.csv', encoding = 'utf-8')
dff
```

	cik	year	name	item_1_text	gvkey
0	1041588	2020	ACCESS-POWER INC	fixed expenses are previosuly documented in an...	66119
1	315374	2020	HURCO COMPANIES INC	General Hurco Companies, Inc. is an internatio...	5788
2	1622996	2020	ACRO BIOMEDICAL CO., LTD.	We have been engaged in the business of develo...	27584
3	1191334	2020	Chun Can Capital Group	CORPORATE HISTORY Chun Can Capital Group (form...	153614
4	1191334	2020	Chun Can Capital Group	CORPORATE HISTORY Chun Can Capital Group (form...	153614
...	...	...	...	...	...
5476	740664	2020	R F INDUSTRIES LTD	General RF Industries, Ltd. (together with sub...	2829
5477	1074828	2020	KNOW LABS, INC.	BACKGROUND AND CAPITAL STRUCTURE Know Labs, In...	166430
5478	40570	2020	GEE Group Inc.	General GEE Group Inc. (the Company , us , ...	5050
5479	1341726	2020	GULFSLOPE ENERGY, INC.	General GulfSlope Energy, Inc. is an independe...	175595
5480	72633	2020	NORTH EUROPEAN OIL ROYALTY TRUST	(a) General Development of Business. North Eur...	7959

5481 rows × 5 columns

- Define an user-defined function to convert the text to lowercase, remove punctuation, and remove English stopwords. Specifically for the last component, we split the text into individual words first, then only kept the non-stop words and joined them altogether for the final result.

```
translator = str.maketrans('', '', string.punctuation)
sw = stopwords.words('english')

def clean_text(text):
    ''' This function takes a string as input and
        returns a cleaned version of the string
        Specifically, it makes the string into lower case and remove punctuations
    '''
    text_lower = text.lower() # make it lowercase
    text_no_punctuation = text_lower.translate(translator) # remove punctuation
    clean_words = [w for w in text_no_punctuation.split() if w not in sw] # remove stopwords
    return ' '.join(clean_words)
```

- Apply the function to the loaded dataset.

```
dff['text1_clean'] = dff['item_1_text'].apply(clean_text)
```

After processing, we stored the result in a new column called *"text1_clean"* that contains the cleaned and standardized text for each firm's 10-K item 1 description. Specifically, this text is all lowercased, punctuation-free, and stopword-free, which makes it suitable for downstream natural language processing (NLP) tasks such as frequency analysis, topic modeling, or building machine learning classifiers.

Sample result:

Before cleaning (left)

```
0       fixed expenses are previosuly documented in an...
1       General Hurco Companies, Inc. is an internatio...
2       We have been engaged in the business of develo...
3       CORPORATE HISTORY Chun Can Capital Group (form...
4       CORPORATE HISTORY Chun Can Capital Group (form...
                                ...
5476    General RF Industries, Ltd. (together with sub...
5477    BACKGROUND AND CAPITAL STRUCTURE Know Labs, In...
5478    General GEE Group Inc. (the  Company ,  us ,  ...
5479    General GulfSlope Energy, Inc. is an independe...
5480    (a) General Development of Business. North Eur...
```

After cleaning (right)

```
[nltk_data]   Package stopwords is already up-to-date!
0       fixed expenses previosuly documented 8k 235000...
1       general hurco companies inc international indu...
2       engaged business developing marketing products...
3       corporate history chun capital group formerly ...
4       corporate history chun capital group formerly ...
                                ...
5476    general rf industries ltd together subsidiarie...
5477    background capital structure know labs inc inc...
5478    general gee group inc company us incorporated ...
5479    general gulfslope energy inc independent crude...
5480    general development business north european oi...
```

b. Keyword Analysis

We conducted the keyword analysis with the cleaned format of the item 1 text from the 10-K report with the following steps:

Step 1. Data Preparation – Filtering Firms in Selected Industry

First, we first merged the *10-K data* with the *public_firms.csv* dataset using the *gvkey* value as the key, so that all companies are matched to their corresponding item 1 texts.

In this step, we found 3 duplicative rows in the *2020_10K_item1_full.csv* sharing the same item 1 text entries for ERA GROUP INC. According to the *public_firms.csv* source of truth, ERA's *gvkey* value should be 187766. However, one of the three rows has a *gvkey* value of '8092', which actually belongs to BRISTOW GROUP INC. This could happen due to the fact that BRISTOW GROUP INC completed a merger with ERA GROUP INC. in 2020, so the two companies' *gvkey* codes may have been used interchangeably (Bristow Group Inc., 2020).

```
#check for duplications

dff[dff['gvkey'].isin([8092, 187766])]
```

✓ 0.0s

	cik	year	name	item_1_text	gvkey	text1_clean
2357	1525221	2020	ERA GROUP INC.	Unless the context indicates otherwise, the te...	8092	unless context indicates otherwise terms us co...
2358	1525221	2020	ERA GROUP INC.	Unless the context indicates otherwise, the te...	187766	unless context indicates otherwise terms us co...
2359	1525221	2020	ERA GROUP INC.	Unless the context indicates otherwise, the te...	187766	unless context indicates otherwise terms us co...

Thus, we chose to remove:

- Row index [2357] as its gvkey value does not match the record in public_firms.csv to ensure consistency by retaining only the record with gvkey '187766' for ERA GROUP INC.
- Row index [2358] to remove redundancy.

To remove these duplicated rows, we used the *drop()* method to delete 2 rows whose indexes are 2357 and 2358. As a result, the deduplicated data set only had 5479 rows left versus 5481 rows before.

```
#remove duplicated rows

dff_c = dff.drop(dff.index[[2357, 2358]])
dff_c
```

[414]

	cik	year	name	item_1_text	gvkey	text1_clean
0	1041588	2020	ACCESS-POWER INC	fixed expenses are previosuly documented in an...	66119	fixed expenses previosuly documented 8k 235000...
1	315374	2020	HURCO COMPANIES INC	General Hurco Companies, Inc. is an internatio...	5788	general hurco companies inc international indu...
2	1622996	2020	ACRO BIOMEDICAL CO., LTD.	We have been engaged in the business of develo...	27584	engaged business developing marketing products...
3	1191334	2020	Chun Can Capital Group	CORPORATE HISTORY Chun Can Capital Group (form...	153614	corporate history chun capital group formerly ...
4	1191334	2020	Chun Can Capital Group	CORPORATE HISTORY Chun Can Capital Group (form...	153614	corporate history chun capital group formerly ...
...	...	...	...	...	...	...
5476	740664	2020	R F INDUSTRIES LTD	General RF Industries, Ltd. (together with sub...	2829	general rf industries ltd together subsidiarie...
5477	1074828	2020	KNOW LABS, INC.	BACKGROUND AND CAPITAL STRUCTURE Know Labs, In...	166430	background capital structure know labs inc inc...
5478	40570	2020	GEE Group Inc.	General GEE Group Inc. (the Company , us , ...	5050	general gee group inc company us incorporated ...
5479	1341726	2020	GULFSLOPE ENERGY, INC.	General GulfSlope Energy, Inc. is an independe...	175595	general gulfslope energy inc independent crude...
5480	72633	2020	NORTH EUROPEAN OIL ROYALTY TRUST	(a) General Development of Business. North Eur...	7959	general development business north european oi...

5479 rows × 6 columns

Then, we ultimately merged this deduplicated 10-K data with the public_firm dataset.

```
#part D1: merge two dataframes on gvkey

merged_df = dff_c.merge(company_data, left_on = "gvkey", right_on = "gvkey", how = "inner")
merged_df
```

Python

	cik	year	name	item_1_text	gvkey	text1_clean	fyear	location	conm	ipodate	sic	prcc_c	ch	ni	asset	sale	roa	market_share
0	92380	2020	SOUTHWEST AIRLINES CO	Company Overview Southwest Airlines Co. (the ...	9882	company overview southwest airlines co company...	1995	USA	SOUTHWEST AIRLINES	NaN	4512	23.0000	317.363	182.626	3256.122	2872.751	0.056087	0.032177
1	92380	2020	SOUTHWEST AIRLINES CO	Company Overview Southwest Airlines Co. (the ...	9882	company overview southwest airlines co company...	1996	USA	SOUTHWEST AIRLINES	NaN	4512	22.0000	581.841	207.337	3723.479	3406.170	0.055684	0.038152
2	92380	2020	SOUTHWEST AIRLINES CO	Company Overview Southwest Airlines Co. (the ...	9882	company overview southwest airlines co company...	1997	USA	SOUTHWEST AIRLINES	NaN	4512	24.6250	623.343	317.772	4246.160	3816.821	0.074838	0.042751
3	92380	2020	SOUTHWEST AIRLINES CO	Company Overview Southwest Airlines Co. (the ...	9882	company overview southwest airlines co company...	1998	USA	SOUTHWEST AIRLINES	NaN	4512	22.6875	378.511	433.431	4715.996	4163.980	0.091907	0.046640
4	92380	2020	SOUTHWEST AIRLINES CO	Company Overview Southwest Airlines Co. (the ...	9882	company overview southwest airlines co company...	1999	USA	SOUTHWEST AIRLINES	NaN	4512	16.1250	418.819	474.378	5652.113	4735.587	0.083929	0.053042
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
374	1048911	2020	FEDEX CORP	Overview FedEx Corporation (FedEx) was incor...	4598	overview fedex corporation fedex incorporated ...	2016	USA	FEDEX CORP	NaN	4513	186.2000	3969.000	2997.000	48552.000	60319.000	0.061728	0.675616
375	1048911	2020	FEDEX CORP	Overview FedEx Corporation (FedEx) was incor...	4598	overview fedex corporation fedex incorporated ...	2017	USA	FEDEX CORP	NaN	4513	249.5400	3265.000	4572.000	52330.000	65450.000	0.087369	0.733087
376	1048911	2020	FEDEX CORP	Overview FedEx Corporation (FedEx) was incor...	4598	overview fedex corporation fedex incorporated ...	2018	USA	FEDEX CORP	NaN	4513	161.3300	2319.000	540.000	54403.000	69693.000	0.009926	0.780611
377	1048911	2020	FEDEX CORP	Overview FedEx Corporation (FedEx) was incor...	4598	overview fedex corporation fedex incorporated ...	2019	USA	FEDEX CORP	NaN	4513	151.2100	4881.000	1286.000	73537.000	69217.000	0.017488	0.775280
378	1048911	2020	FEDEX CORP	Overview FedEx Corporation (FedEx) was incor...	4598	overview fedex corporation fedex incorporated ...	2020	USA	FEDEX CORP	NaN	4513	259.6200	7087.000	5231.000	82777.000	83959.000	0.063194	0.940401

379 rows × 18 columns

Step 2. Keyword Extraction – Word Counts and TF-IDF

In this step, we used two different methods to generate the top 10 keywords for each company:

- **Word counts**: By counting their frequency in cleaned item 1 text, this method gets the top 10 most frequent keywords and appends them to an accumulator.

```
                  conm                        top_10_keywords_wordcounts
0    SOUTHWEST AIRLINES  company southwest aircraft flight also service...
1    SOUTHWEST AIRLINES  company southwest aircraft flight also service...
2    SOUTHWEST AIRLINES  company southwest aircraft flight also service...
3    SOUTHWEST AIRLINES  company southwest aircraft flight also service...
4    SOUTHWEST AIRLINES  company southwest aircraft flight also service...
..                 ...                                                ...
374          FEDEX CORP  fedex express us service services freight grou...
375          FEDEX CORP  fedex express us service services freight grou...
376          FEDEX CORP  fedex express us service services freight grou...
377          FEDEX CORP  fedex express us service services freight grou...
378          FEDEX CORP  fedex express us service services freight grou...

[379 rows x 2 columns]
```

- **TF-IDF**: By importing TfidfVectorizer from scikit-learn library in Python, this method gets the top 10 keywords for each document based on their TF-IDF score. This is achieved via a user-defined function that takes the document list as inputs and returns the relevant keywords (shown as below).

```python
from sklearn.feature_extraction.text import TfidfVectorizer

def get_keywords_tfidf(document_list):
    '''
    This function gets a list of documents as input and returns a list of top 10 keywords for each document using TF-IDF scores.
    Input: A list of documents (text)
    Output: The corresponding top 10 keywords for each document based on tf-idf values
    '''
    vectorizer = TfidfVectorizer() # Step 1: Create a TF-IDF vectorizer
    tfidf_matrix = vectorizer.fit_transform(document_list) # Step 2: Calculate the TF-IDF matrix
    feature_names = vectorizer.get_feature_names_out() # Step 3: Get feature names (words)

    # Step 4: Extract top 10 keywords for each document
    top_keywords = [] # accumulator
    for i in range(len(document_list)):
        feature_index = tfidf_matrix[i, :].nonzero()[1]
        feature_value = [tfidf_matrix[i, x] for x in feature_index]
        tfidf_scores = zip(feature_index, feature_value)
        sorted_tfidf_scores = sorted(tfidf_scores, key=lambda x: x[1], reverse=True)
        top_keywords.append(' '.join([feature_names[i] for i, _ in sorted_tfidf_scores[:10]]))

        if i % 200 == 199:
            print(f'Processed {i+1}/{len(document_list)} documents.')

    return top_keywords

docs = merged_df['text1_clean'].tolist()
len(docs)

# Training the docs to get TD-IDF scores
tfidf_keywords = get_keywords_tfidf(docs)

# Add a new column in the dataframe
merged_df['top_keyword_tfidf'] = tfidf_keywords
```

Sample result:

```
  'company southwest fare aircraft flight airline also alist max airlines',
 ...
  'fedex express freight shipping us delivery tnt ground service logistics',
```

Step 3. Visualization with Word Clouds

To visualize the results across all firms within the Transportation by Air sector, we aggregated the keyword terms and generated two word clouds:

- **Word Count Word Cloud** → This shows the most frequent business terms in the sector.
- **TF-IDF Word Cloud** → This Highlights the distinctive terms that uniquely characterize firms in the sector.

In order to do this, we leveraged the WordCloud and Matplotlib libraries. First, we merged the top 10 keywords generated together for each method, then we used WordCloud to visualize them with custom dimensions e.g., width, height, font size, color.

```python
from wordcloud import WordCloud
import matplotlib.pyplot as plt

# prepare text
text_wordcounts = ' '.join(merged_unique['top_10_keywords_wordcounts'].tolist())
text_tfidf = ' '.join(merged_unique['top_keyword_tfidf'].tolist())

# return result: wordcounts method
wordcloud1 = WordCloud(width=800, height=400, max_font_size=100, background_color='white').generate(text_wordcounts) # note that text is a string, not a list
```

Then, we used Matplotlib to plot and save the images (see results below).

```
plt.figure(figsize=(10,5))
plt.axis('off')
plt.imshow(wordcloud1)
plt.savefig('keyword_wordcounts.png') # save as PNG file
plt.show()

# return result: TFIDF method
wordcloud2 = WordCloud(width=800, height=400, max_font_size=100, background_color='white').generate(text_tfidf) # note that text is a string, not a list
plt.figure(figsize=(10,5))
plt.axis('off')
plt.imshow(wordcloud2)
plt.savefig('keyword_tfidf.png') # save as PNG file
plt.show()
```

- Result images:

Generated by WordCounts method:

Generated by TF-IDF method:

c. Word Embedding

Step 1. Training the Word2Vec Model

Using the cleaned text from the 10-K Item 1 dataset, we trained a Word2Vec model to capture semantic relationships among words in Transportation by Air industry's 10-K filings. The model was set up with the following parameters:

- **vector_size**: Number of dimensions of the word embeddings (default: **100)**
- **window**: Maximum distance between a target word and its surrounding context words (default: **5**)
- **min_count**: Minimum frequency threshold for words to be included in training. Words that occur fewer times than this threshold are ignored (default: **5**)
- **workers**: Number of parallel worker threads used during training (default: **3**)
- **sg**: The training algorithm. If set to **0**, the model uses the **Continuous Bag of Words (CBOW)** method; if set to **1**, it uses the **Skip-gram** method. The default training algorithm is **CBOW (sg=0)**.

Then we split the cleaned item 1 text into lists of lists of words for model training.

```
# list of list of words
docs2 = [row.split() for row in dff['text1_clean']]

# training word2vec model using the list of words above
model = Word2Vec(docs2, min_count=5, vector_size=50, workers=3, window=5, sg = 1)

# save the model for future use
model.save("word2vec_final.model")

# load model from stored file
model = Word2Vec.load("word2vec_final.model")
```

Step 2. Representative Keyword Selection and Word2Vec Similarity Analysis

From the two WordClouds generated above, we then inspected and narrowed down three representative industry-relevant keywords.

- **<u>Airline</u>**: The similarity analysis shows that the model understands **"airline"** not just as a standalone term, but as part of a **network of related concepts**: customer services (passengers), operations (flights, inflight, aircraft), and industry ecosystem (forwarders, aviation, gogo).

```
# find the most relevant five words for "airline"

model.wv.most_similar('airline')
```

```
[('airlines', 0.8820410966873169),
 ('passengers', 0.8358259201049805),
 ('passenger', 0.8145914077758789),
 ('flights', 0.7962758541107178),
 ('flight', 0.7910593748092651),
 ('inflight', 0.7899420857429504),
 ('aircraft', 0.7713428139686584),
 ('freight', 0.7637747526168823),
 ('cruise', 0.7588531970977783),
 ('travel', 0.7581068873405457)]
```

- **<u>Aircraft</u>**: The similarity network around **"aircraft"** demonstrates how the Word2Vec model categorized its related terms into **sub-domains of aviation**, namely commercial transport (e.g., passenger, Airbus, Boeing), technical design (e.g., airframe, turboprop, jets), and defense (e.g., unmanned, fighter).

```
# find the most relevant five words for "aircraft"

model.wv.most_similar('aircraft')
```

```
[('passenger', 0.8449004888534546),
 ('airframe', 0.8300570845603943),
 ('boeing', 0.8182541131973267),
 ('unmanned', 0.8128969073295593),
 ('flight', 0.8121019601821899),
 ('airbus', 0.8076152801513672),
 ('turboprop', 0.8023190498352051),
 ('jets', 0.797966718673706),
 ('widebody', 0.7977890968322754),
 ('airliners', 0.797055184841156)]
```

- **<u>Fedex</u>**: The similar results for **"fedex"** marks that the Word2Vec model effectively captures the **logistics ecosystem**, linking freight forwarding not only with technical terms (*freight, forwarding*) but also with **major companies** *(DHL, TNT)*, **transport modes** *(airlines)*, and **retail logistics** *(Nordstrom, pickup)*.

```
# find the most relevant five words for "fedex"

model.wv.most_similar('fedex')
```

```
[('airlines', 0.8007073998451233),
 ('freight', 0.7794128060340881),
 ('forwarding', 0.7776100039482117),
 ('usps', 0.7636470198631287),
 ('dhl', 0.756200909614563),
 ('airline', 0.7550895810127258),
 ('pickup', 0.7525520920753479),
 ('nordstrom', 0.7449009418487549),
 ('forwarder', 0.7389336824417114),
 ('tnt', 0.7368916273117065)]
```

4. Comprehensive firm analysis

In this section, we decided to analyze the market status of ***American Airlines Inc.***, one of the largest U.S carriers with a global network of passengers and fly teams. Our analysis unfolds in three main parts. First, we mapped out the closest competitors to American Airlines. Second, we benchmarked the general performance across this competitive landscape. Lastly, we conducted an in-depth evaluation of American Airlines and provided our recommendations to the firm.

Part 1: Competitor identification

We converted the top keywords extracted for each firm earlier into word embeddings with the word2vec model trained in the Text cleaning and NLP analysis section. We added up the embeddings for each firm to create the firm-level embeddings, then used it to find the focal firm's most similar competitors. This methodology allows us to extract a list of competitors that are more likely to share the business activities and focused areas, as described in their 10-K filings, with American Airlines than relying solely on SEC sector classification. This method also helps identify the subtle nuances among different air transport sub-sectors, for instance, differentiating passenger carriers from cargo specialists and service providers, thus arriving at a more accurate list of competitors:

- HAWAIIAN HOLDINGS INC

- SPIRIT AIRLINES INC
- DELTA AIR LINES INC
- ALLEGIANT TRAVEL CO
- SOUTHWEST AIRLINES
- AIR TRANSPORT SERVICES GROUP
- HARBOR DIVERSIFIED INC
- JETBLUE AIRWAYS CORP
- FEDEX CORP
- ATLAS AIR WORLDWIDE HLDG INC

```
# part 1: finding competitors

from DocumentSimilarity import DocumentSimilarity

# Create an instance
docsim = DocumentSimilarity(model = model, gvkeys=merged_unique['gvkey'], conm = merged_unique['conm'],
                            keywordslist = merged_unique['top_keyword_tfidf'])

# Get firm's embedding
docsim.get_firm_embedding(firm = 1388)                    # AMERICAN AIRLINES INC

# Get most similar firms based on word2vec embeddings
docsim.most_similar(firm = 1388, topn = 10)
```

Part 2: Competitive landscape benchmark

a. **Revenue**

Here we will look at the absolute revenue of American Airlines and its competitors in 2020 and their revenue growth over time. Since the industry is cyclical and year-to-year revenue can be distorted by shocks (e.g., COVID), we examine these companies' revenue Compounded Annual Growth Rate (CAGR). CAGR is calculated as:

$$\left(\frac{\text{Final revenue}}{\text{Initial revenue}}\right)^{\frac{1}{\text{Time period}}} - 1$$

This measure was chosen over average year-over-year percentage change to smoothen volatility and accurately capture firms' long-term growth trajectory. To compute the revenue CAGR, we created a user-defined function called *cagr()* that invokes the usual growth formula on a start value, end value, and the time interval expressed in years. Instead of computing growth row-wise within the data frame, we did it firm-wise. We retrieved the initial and terminal non-zero revenue values, subtracted the start and end year to get the time interval, and invoked the *cagr()* function. The results for each firm were saved in a dictionary and eventually converted to a data frame for data comparing and sorting.

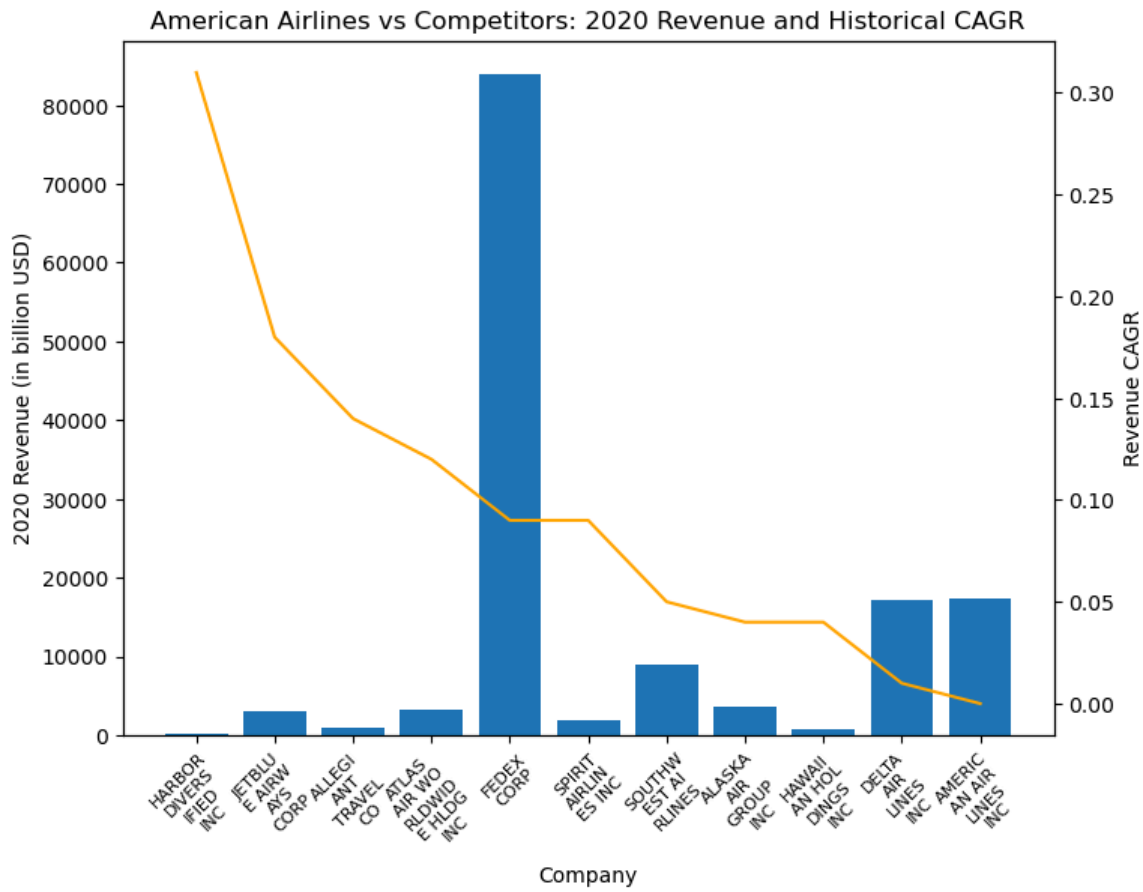
```
def cagr(start_year, end_year, duration):
    return (round((end_year / start_year) ** (1 / duration) - 1, 2))

competitor_cagr = {'company': [], 'cagr': [], 'duration' : []}


for name, data in competitor_df.groupby('conm'):
    data = data.sort_values('fyear')

    # extract start year sales by going through non-zero sales and taking 1st row of each group data
    start_year = data.loc[data['sale'] != 0].iloc[0]['sale']

    # extract end year sales by going through non-zero sales and taking last row of each group data
    end_year = data.loc[data['sale'] != 0].iloc[-1]['sale']

    ## note: i notice Saker has a start year sale of 0, which causes an error in cagr calculation.
    ## so I added the != condition to only select first and last non-zero values.


    # calculate duration
    duration = data.iloc[-1]['fyear'] - data.iloc[0]['fyear']

    #c calculate cagr
    firm_cagr = cagr(start_year, end_year, duration)

    #append results to dictionary
    competitor_cagr['company'].append(name)
    competitor_cagr['cagr'].append(firm_cagr)
    competitor_cagr['duration'].append(duration)

competitor_cagr_df = pd.DataFrame(competitor_cagr, columns=['company', 'cagr', 'duration']).sort_values('cagr', ascending=False)
competitor_cagr_df
```

To visualize this better, we used the *matplot* and *textwrap* libraries to plot these companies' 2020 revenue and historical CAGR on a line-bar chart, with the left axis representing the 2020 revenues and the right axis representing the CAGRs.

Overall, bigger companies like FedEx, American Airlines, and Delta tend to have slower revenue growth. Their size, establishment, and broad revenue streams may pose challenges to sustain high CAGRs due to both saturation and competition from emerging low cost carriers. Taking the leadership was FedEx, which outplayed other companies with ~$84B revenue in 2020 and a steady

long-term growth of ~9% CAGR for over 26 years. Next, the traditional airlines such as American Airlines ($17.3B) and Delta ($17.1B) also featured the top revenues in 2020, yet these businesses showed stagnation over 25 years with flat, almost-zero historical CAGRs. Southwest and Alaska remain mid-tier carriers with more modest revenues, $9.0B and $3.6B respectively and marginal expansion with ~4-5% CAGR only. Smaller, niche airline carriers stood out with more outstanding growth, namely Harbor Diversified (31%), JetBlue (~18%), Allegiant (~14%), and Atlas Air (12%).

```
import textwrap

competitor_rev_combined = competitor_cagr_df.merge(rev_2020, left_on = "company", right_on = "conm", how = "inner")
competitor_rev_combined

# Create a figure and a primary axes object
fig, ax1 = plt.subplots(figsize=(8, 6))

# Plot the bar chart on the primary axes
ax1.bar(range(len(competitor_rev_combined['company'])), competitor_rev_combined['sale'])
ax1.set_xlabel('Company')
ax1.set_ylabel('2020 Revenue (in million USD)')

# Get the labels from the 'company' column
raw_labels = competitor_rev_combined['company'].tolist()


# Wrap each label at 5 characters
wrapped_labels = [textwrap.fill(label,6) for label in raw_labels]

# Apply wrapped labels using plt.xticks()
plt.xticks(ticks=range(len(competitor_rev_combined['company'])), labels=wrapped_labels, rotation=0, fontstyle = 'italic', fontsize = 8)

# Create a secondary axes object that shares the same x-axis
ax2 = ax1.twinx()

# Plot the line chart on the secondary axes
ax2.plot(range(len(competitor_rev_combined['company'])), competitor_rev_combined['cagr'], color='orange')
ax2.set_ylabel('Revenue CAGR')

# Add a title and legends
plt.title('American Airlines vs Competitors: 2020 Revenue and Historical CAGR')
plt.show()
```



b. Profitability & Efficiency

ROA

To begin with, we looked at the selected companies' 2020 ROA and compared them against the Airline Industry Benchmark to identify outperformers and laggards. Following the same approach, we filtered for data of only American Airlines and its competitors in 2020 and sorted their ROAs by descending order.

```
# 2020 ROA

competitor_df = df[df["gvkey"].isin(company_code)]
roa_2020 = competitor_df[competitor_df['fyear'] == 2020].sort_values('roa', ascending=False)
roa_2020[['conm', 'roa']]
```

In order to find a reasonable benchmark for the sector's typical performance, we studied the mean and median of all Transportation by Air companies' ROA.

Since the mean (-0.078) is less than the median (-0.106), it suggests a left-skewed distribution in 2020 where few players posted extremely poor results

that dragged the entire sector down. Hence, we would recommend using the median ROA to benchmark the industry's health..

	conm	roa
124170	HARBOR DIVERSIFIED INC	0.113462
14369	FEDEX CORP	0.063194
119487	ATLAS AIR WORLDWIDE HLDG INC	0.059757
178946	AIR TRANSPORT SERVICES GROUP	0.010699
201501	SPIRIT AIRLINES INC	-0.051043
190505	ALLEGIANT TRAVEL CO	-0.056489
36248	SOUTHWEST AIRLINES	-0.088875
174886	JETBLUE AIRWAYS CORP	-0.101000
1442	AMERICAN AIRLINES INC	-0.122083
118091	HAWAIIAN HOLDINGS INC	-0.128439
11302	DELTA AIR LINES INC	-0.172023

The result showed us that the COVID pandemic's lingering effect in 2020 hit passenger carriers the hardest. No matter if they are legacy or low-cost players or not, this group underscored with a negative ROAs that are mostly lower than industry median. This also captured the weak travelling demand associated with negative profits and low utilization of assets. On the contrary, cargo operators like FedEx, Atlas Air Worldwide Holding Inc, Harbor Diversified, and Air Transport Services Group achieved positive ROAs. This result may be credit to the e-commerce boom and surging demand for contactless shipping. Also, most of these companies established long-term contracts or partnerships with e-commerce and delivery giants like Amazon, DHL, and UPS, which helped shield them from volatility.

At the same time, we were also interested in seeing how these companies' ROA evolved over years. Therefore, we also created a snapshot of their historical average ROA, using *mean()* function.

```
# historical average ROA

competitor_df.groupby('conm')['roa'].mean().sort_values(ascending=False)
```

The result categorized the profitability performance of these firms into 2 main groups. On one side, low cost carriers such as Spirit, Allegiant, Southwest, Alaska are remarkable for their steadily positive ROA averages. On the other side, legacy players such as American Airlines, Delta, and Hawaiian show weak or even negative averages, which implies structural profit issues. Simultaneously, there seems to showcase a general theme: while passenger airlines are cyclical and crisis-loss prone, cargo and low-cost airline operators tend to maintain healthier returns in the long run, which highlights the significance of diversifying business models.

```
conm
SPIRIT AIRLINES INC                0.104676
ALLEGIANT TRAVEL CO                0.082701
AIR T INC                          0.062521
SOUTHWEST AIRLINES                 0.051203
ALASKA AIR GROUP INC               0.030829
ATLAS AIR WORLDWIDE HLDG INC       0.029895
AMERICAN AIRLINES INC             -0.010141
DELTA AIR LINES INC               -0.011284
AIR TRANSPORT SERVICES GROUP      -0.027134
SAKER AVIATION SERVICES INC       -0.072054
HAWAIIAN HOLDINGS INC             -0.851615
```

Net income margin

We closed the profitability analysis by reviewing the bottom-line margin of these companies. We looked into their 2020 net margin and historical average net income margin. The same pattern emerged: legacy airlines such as American (-48.7%), Delta (-72.4%), and Hawaiian (-60.5%) suffered the steepest losses in

2020, a testament to how capital-intensive companies respond to demand shocks. Historically, their margins have also been behind peers, with American (-2.5%) and Delta (-4.2%), showing chronic profitability issues. While 2020 is a disappointing financial year for low cost and niche carriers, they still remain highly resilient in the long term, with Southwest, Spirit and Allegiant earning profit at a rate of 5.5%, 7.6%, and 8.2% respectively. Meanwhile, cargo operators are the only group with solid performance in both 2020 and historically. Specifically, Harbor (21.4%), Atlas (11.2%), FedEx (6.2%), and ATSG (2.0%) all achieved positive 2020 margins and their long-term trend of 2-5% also verifies this robustness. This is in line with above ROA observations: cargos were insulated from volatile demand by e-commerce surge and multi-year service contracts.

	conm	ni_margin
124170	HARBOR DIVERSIFIED INC	0.213838
119487	ATLAS AIR WORLDWIDE HLDG INC	0.112200
14369	FEDEX CORP	0.062304
178946	AIR TRANSPORT SERVICES GROUP	0.020448
190505	ALLEGIANT TRAVEL CO	-0.185939
201501	SPIRIT AIRLINES INC	-0.236848
36248	SOUTHWEST AIRLINES	-0.339744
174886	JETBLUE AIRWAYS CORP	-0.457897
1442	AMERICAN AIRLINES INC	-0.487453
118091	HAWAIIAN HOLDINGS INC	-0.604791
11302	DELTA AIR LINES INC	-0.724481

Historical average net income margin:

```
conm
ALLEGIANT TRAVEL CO                0.081605
SPIRIT AIRLINES INC                0.075687
SOUTHWEST AIRLINES                 0.055137
ATLAS AIR WORLDWIDE HLDG INC       0.054185
FEDEX CORP                         0.036902
JETBLUE AIRWAYS CORP               0.015482
AIR TRANSPORT SERVICES GROUP       0.010712
AMERICAN AIRLINES INC             -0.025435
DELTA AIR LINES INC               -0.041907
HARBOR DIVERSIFIED INC                 -inf
HAWAIIAN HOLDINGS INC                  -inf
```

Note: Harbor Diversified Inc and Hawaiian Holdings Inc have infinite margin records due to multiple years with zero-reported sales.

c. Market Position

We defined a firm's market share as its annual sales divided by the total annual sales of all firms in the air transportation industry. After filtering out for 2020 data only and computing the calculations for all sectors' firms, we then extracted and ranked only the results from American and their shortlisted competitors.

```
#2020 Market share

# we define it as Firm's Sales / Total Sales of all Air Transportation industry

company_data_2020 = company_data[company_data['fyear'] == 2020]
company_data_2020['market_share'] = company_data_2020['sale'] / (company_data_2020['sale'].sum()) * 100

competitor_share_2020 = company_data_2020[company_data_2020["gvkey"].isin(company_code)].sort_values('market_share', ascending=False)
competitor_share_2020
```

Market Share of Top Players (by absolute values)

The result below shows FedEx operated in a different segment from other first (logistics/courier) and emerged as the dominant and financially robust winner with a

market share of 30.63%. Large passenger airlines (e.g. American Airlines, Delta Air Lines) have market shares between 3% and 6%, which indicates that high sales (and market share) did not translate to profitability in 2020, when the COVID-19 pandemic fundamentally crippled the entire air transportation sector.

	conm	market_share
14369	FEDEX CORP	30.627201
1442	AMERICAN AIRLINES INC	6.323593
11302	DELTA AIR LINES INC	6.236044
36248	SOUTHWEST AIRLINES	3.300598
119487	ATLAS AIR WORLDWIDE HLDG INC	1.171375
174886	JETBLUE AIRWAYS CORP	1.078677
201501	SPIRIT AIRLINES INC	0.660274
178946	AIR TRANSPORT SERVICES GROUP	0.572926
190505	ALLEGIANT TRAVEL CO	0.361166
118091	HAWAIIAN HOLDINGS INC	0.308177
124170	HARBOR DIVERSIFIED INC	0.067832

Percent Change in Market Share Overtime

In this part, we derived each firm's percentage change in market share by subtracting its starting available year's market share from its ending available year's share value. The results were also viewed in parallel with the time period to disclose how the market share had shifted over each period observed for firms.

```python
for name, data in competitor_df.groupby('conm'):
    data = data.sort_values('fyear')

    # get start year market share = start year sales / total sales of all same-industry firms in that year
    start_year = data.iloc[0]['fyear']
    start_share = (data.iloc[0]['sale'] / company_data[company_data['fyear'] == start_year]['sale'].sum()) * 100

    # get end year sales
    end_year = data.iloc[-1]['fyear']
    end_share = (data.iloc[-1]['sale'] / company_data[company_data['fyear'] == end_year]['sale'].sum()) * 100

    # calculate duration
    duration = data.iloc[-1]['fyear'] - data.iloc[0]['fyear']

    #c calculate market share percent point change
    share_change = round(end_share - start_share, 5)

    #append results to dictionary
    market_share_change['company'].append(name)
    market_share_change['market_share_percent_point_change'].append(share_change)
    market_share_change['duration'].append(duration)
```

American Airlines's sharp decline of -3.1% market share despite over 25 years in the industry. Conversely, Southwest and Atlas Air Worldwide solidified their positions as consistently growing players over the same period, with 1.57% and 1.07% share increase respectively. Firms with shorter duration observed like Spirit also showed rapid growth with 0.45% increase over 11 years, demonstrating how agile and line operating models can be effective in gaining a foothold in the market. Resilient cargo and low-cost carriers are the winners with the most substantial growth, such as Atlas Air Worldwide with 1.07%, Spirit Airlines with 0.45%. This strikes the strategic importance of air cargo and streamlined business models in contributing to overall industry dynamics.

	company	market_share_percent_point_change	duration
9	SOUTHWEST AIRLINES	1.56630	25
5	ATLAS AIR WORLDWIDE HLDG INC	1.06798	25
10	SPIRIT AIRLINES INC	0.44545	11
2	ALASKA AIR GROUP INC	0.44508	25
3	ALLEGIANT TRAVEL CO	0.31812	15
7	HAWAIIAN HOLDINGS INC	0.09875	25
1	AIR TRANSPORT SERVICES GROUP	0.05181	19
8	SAKER AVIATION SERVICES INC	0.00128	20
0	AIR T INC	-0.01128	26
6	DELTA AIR LINES INC	-1.12556	25
4	AMERICAN AIRLINES INC	-3.10028	25

In conclusion, in 2020, the U.S. air transportation industry experienced a redistribution of market share. Southwest and Atlas Air emerged as clear winners, while American Airlines suffered the steepest decline. This highlights a structural resilience of low-cost and cargo carriers, compared to traditional network airlines that rely on long-haul international and corporate segments.

Part 3: In-depth analysis for American Airlines Inc.

Historical stock price

Since our company of interest, American Airlines Inc (gvkey: 1388), does not have any historical stock price data, we will use the stock price data of American Airlines Group Inc (gvkey: 1045) for the analysis. This could be due to the fact that American Airlines Group Inc. is the public parent holding company of American Airlines Inc. Also, in 2013, American Airlines merged with US Airways and thus formed American Airlines Group Inc., which has been listed with the 'AAL' ticker ever since (American Airlines, 2013).

First, we plotted American Airline's historical stock prices against years from 1994 to 2020 on a line chart:

```
american_group_data = company_data[company_data['conm'] == "AMERICAN AIRLINES GROUP INC"]

astock = american_group_data[['prcc_c', 'fyear']]
plt.plot(astock.fyear, astock.prcc_c)
plt.grid(True)

plt.title('American Airlines Group Inc: Historical stock prices (in USD)')
plt.xlabel('Year')
plt.ylabel('Stock Price (in USD)')

plt.show()
```

The chart indicates American Airlines Group Inc.'s past stock performance from 1994 through 2020 with extreme volatility sharply peaked in the 1990s and bottomed out around the 2000s, with a long-term plateau and partial recovery in the 2010s, just before another plummet near 2020. The pinnacle the stock price reached was 1997 with $125, and the trough was in 2011 with $2. American's stock began 1995 at around $75 and rose sharply to around $125 in 1997. This implies a period of robust growth, which might have been due to industry development or economic optimism. The significant fall was from 1998 to 2003, which might have occurred due to the increase in fuel prices and the 9/11 incident. The stock declined from 2017 onward and hit around $15–20 by 2020 due to the pandemic impacts.

Historical revenue

Then we also plotted out American Airlines' historical revenue within the same period, by filtering for its fiscal year series and annual sales values:

```
american_data = company_data[company_data['conm'] == "AMERICAN AIRLINES INC"]
asale = american_data[['sale', 'fyear']]
plt.plot(asale.fyear, asale.sale)
plt.grid(True)

plt.title('American Airlines Inc: Historical Revenue (in million USD)')
plt.xlabel('Year')
plt.ylabel('Revenue (in million USD)')

plt.show()
```

Revenue of American Airlines from 1994 until 2007 steadily increased from around $15B to $23B with only negligible dips in the 2001-2002 post-9/11 recession and 2008 financial crisis. From 2013 to 2015, revenue increased from $27B to over $40B, which is potentially the effect from the merger with US Airways to form the world's largest airline at that time. Likewise, revenue took a nosedive sharply from its 2013-2015 peak to around $17–18B, which reflects the significant impact of the COVID-19 pandemic on global air travel. Though revenue exhibits long-term uptrend growth (especially after the merger), it is evident that profitability is not necessarily equivalent to high revenue. High fixed costs of the airline business, high debts, and high vulnerability towards shocks

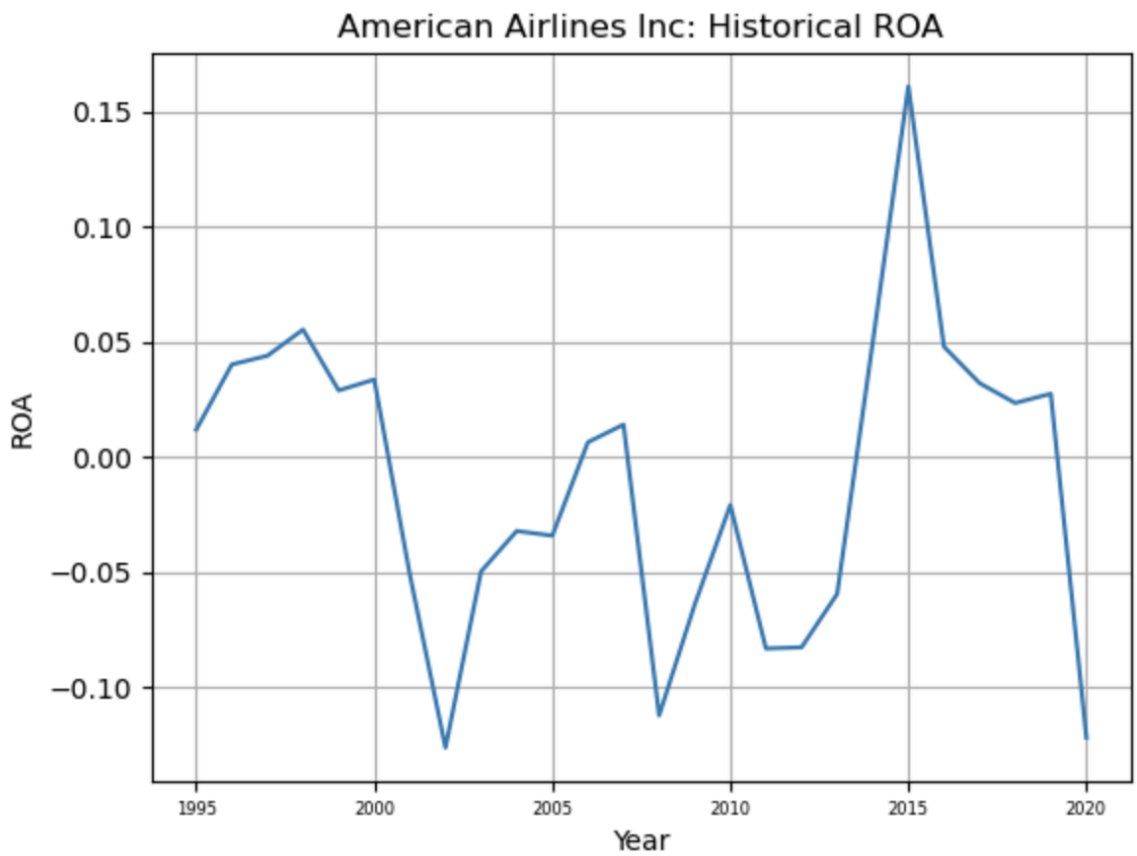
explain why healthy revenue may still result in negative earnings.

Historical ROA

Following the same approach, we plotted out American Airlines' return on assets (ROA) metric over years:

```
american_data = company_data[company_data['conm'] == "AMERICAN AIRLINES INC"]
aroa = american_data[['roa', 'fyear']]
plt.plot(aroa.fyear, aroa.roa)
plt.grid(True)

plt.title('American Airlines Inc: Historical ROA')
plt.xlabel('Year')
plt.ylabel('ROA')

plt.show()
```

The ROA for American Airlines was extremely volatile, often fluctuating between positive and negative values over a long time frame. This is a manifestation of the cyclical and risk-sensitive airline industry that is quickly impacted by fuel price changes, fluctuation of labor cost, and non-price shocks such as economic downturn or global pandemic.



Its biggest declines occurred in 2002–2003 (post-9/11 recession), 2008–2011 (global financial crisis and oil shock), and 2020 (COVID-19 pandemic), all with acute negative ROA. From 2013 to 2015, American Airlines' ROA shot up over 0.15, a spell of elevated efficiency which was propelled by restructuring, industry consolidation, and benign oil prices. While American Airlines experienced periodic spikes in profitability, it consistently lapsed again into adverse ROA, indicating structural profitability problems due to heavy leverage, capital intensity, and responsiveness to shock.

Historical assets

```
aasset = american_data[['asset', 'fyear']]
plt.plot(aasset.fyear, aasset.asset)
plt.grid(True)

plt.title('American Airlines Inc: Historical Assets (in million USD)')
plt.xlabel('Year')
plt.ylabel('Total Assets (in million USD)')

plt.show()
```

Between 1994 to 2012, American's assets remained fairly flat at $18 to 30B, indicating minimal growth and restructuring limitations. Following the 2013 merger with US Airways, total assets doubled by 2015 to $50B and over $70B by 2018 and 2019 as a result of capacity increases and aircraft fleet expansion. Assets rose as high as around $72B by 2019 and 2020 but then declined slightly and remained at historic highs, even as COVID wiped out revenue. This again highlights the capital intensity and high debt leverage within the industry.

When putting both assets and profitability into the picture, we noticed that even as the assets surged, profitability did not keep pace.This implies income and profit could not grow fast enough to sustain that huge expansion of assets and the asset utilization itself is not that effective. Even in 2020, as revenue collapsed, the assets decreased modestly, which implies fixed assets such as aircraft held book value but sat idle during the pandemic period.

Historical net income margin

```
anim = american_data[['asset', 'fyear']]
anim['ni_margin'] = american_data['ni'] / american_data['sale']
plt.plot(anim.fyear, anim.ni_margin)
plt.grid(True)

plt.title('American Airlines Inc: Historical Net Income Margin')
plt.xlabel('Year')
plt.ylabel('Net Income Margin')

plt.show()
```

Compared to its ROA, American Airlines's net income margin is less volatile. This pattern is reasonable as airline ROA often takes in a large asset base, which is subject to change frequently due to new plane accumulation, depreciation and impairment accounting. These changes, however, may not move net income as much. American Airlines realized some modest profitability in the late 1990s, followed by a steep post-9/11 decline and a gradual recovery through 2007. The 2008 financial crisis caused the income margins negative again, but breakeven conditions did return during the period from 2010 to 2014. Profitability peaked in 2015 at nearly 20%, aided by low fuel costs and consolidation, before falling steadily through 2019. In 2020, the margins collapsed to -48%, which was the worst in 25 years due to COVID-19's severe impacts.

Recommendations

In summary, American Airlines has been facing ongoing structural cost pressure. Since 2015, its revenue has continued to rise, but its margins have failed to catch up. Given that, we would like to propose the following recommendations:

1. **Cost Efficiency & Operations**
    - **Upgrading the fleet:** Reduce fuel expenses, replace mature aircraft with more efficient engines
    - **Digital and automation investments:** Streamline operation processes, overhead, as well as optimizing for route profitability

2. **Debt restructuring**
    - **Refinancing:** Pay off older higher-interest loans with cheaper finance if credit has improved
    - **Increase debt maturities:** Avoid near-term cash squeezes by maturing repayment date
    - **Asset sales:** Dispose or spin out non-core assets e.g., used aircraft, stake in loyalty program to release capital

3. **Revenue Mix**
    - Target high-margin segments with special benefits e.g., premium cabin, AAdvantage loyalty program, cargo services

CODE CITATION

1. **startswith():**
   W3Schools. (2025). Python String startswith() Method.
   https://www.w3schools.com/python/ref_string_startswith.asp

2. **groupby(level=0):** For one particular question, we applied the *groupby()* function learned in class to find the number of firms per location. As the data set repeats companies by rows, grouping the data by both location and company name and then using value_counts() would overcount. Thanks to the examples in the *help()* function, we learned that you can group by different levels of a hierarchical index. In our case, grouping the data by the [0] index (location) then counting gives us the precise number of unique firms.

   Code applied:
   unique_loc
   loc_distribution.groupby(level=0).count().sort_values(ascending=False)

```
help(pd.DataFrame.groupby)
✓ 0.0s
```

```
We can groupby different levels of a hierarchical index
using the `level` parameter:

>>> arrays = [['Falcon', 'Falcon', 'Parrot', 'Parrot'],
...           ['Captive', 'Wild', 'Captive', 'Wild']]
>>> index = pd.MultiIndex.from_arrays(arrays, names=('Animal', 'Type'))
>>> df = pd.DataFrame({'Max Speed': [390., 350., 30., 20.]},
...                   index=index)
>>> df
                Max Speed
Animal Type
Falcon Captive      390.0
       Wild         350.0
Parrot Captive       30.0
       Wild          20.0
>>> df.groupby(level=0).mean()
```

3. **Use of .values attribute:** I consulted ChatGPT to debug the following code:

```
result1 = []
for x in stock_2007["conm"]:
if x in stock_2008["conm"]:
        price_08 = stock_2008[stock_2008["conm"] == x]["prcc_c"]
        price_07 = stock_2007[stock_2007["conm"] == x]["prcc_c"]
        percent_drop =  round((price_08 - price_07)/ price_07 * 100, 2)
         result1.append((x, percent_drop))
```

Issue: The ***result1*** list should return the percentage change in 2007-2008 stock prices but it was empty.

Citation:

- Name, version, company of AI tool: Chat GPT-5 mini.

- Chat objective: to debug code

- Use of AI-generated content:

Evaluation of Results: Chat GPT correctly identified the issue, which was because I was doing arithmetic on pandas Series, not real numbers. Hence, when I tried to perform the subtraction of stock prices between 2007 and 2008, Python tried to match these row indexes first. For unmatched cases, it cannot do the subtraction and makes the result empty. That is why we need to apply another way to extract the numbers from the series before doing calculation and ChatGPT suggested adding the *.values* attribute.

Limitations: none.

How I used the AI's response: I added the .values attribute where applicable and the list result finally returned with numerical values.

- URL of chat history:
https://chatgpt.com/share/68e02510-ce3c-8009-aef2-5a553079ff42

- Time and date of chat: Oct 3 12:30 pm

4. **Initiating data frame:** We learned how to use the pd.DataFrame function in class to convert a dictionary to data frame format. For the two following parts of the

project, we apply this methodology to form a new data frame table from a new calculated result, including the definition of new column names. This is learned by consulting the *help* function, with regards to pd.DataFrame:

Codes applied:

- df_result = pd.DataFrame(result, columns=["company", "percent drop"])
- competitor_cagr_df = pd.DataFrame(competitor_cagr, columns=['company', 'cagr', 'duration']).sort_values('cagr', ascending=False)

```
help(pd.DataFrame)
✓ 0.1s
...
 |          * ``None``: No fill restriction.
 |          * 'inside': Only fill NaNs surrounded by valid values
 |            (interpolate).
 |          * 'outside': Only fill NaNs outside valid values (extrapolate).
 |
```

```
Examples
--------
Copy the contents of a DataFrame to the clipboard.

>>> df = pd.DataFrame([[1, 2, 3], [4, 5, 6]], columns=['A', 'B', 'C'])

>>> df.to_clipboard(sep=',')  # doctest: +SKIP
... # Wrote the following to the system clipboard:
... # ,A,B,C
... # 0,1,2,3
... # 1,4,5,6
```

5. **Inner join:**

   W3Schools. (2025). Pandas merge method. W3Schools.https://www.w3schools.com/python/pandas/ref_df_merge.asp

6. **isin():**
   W3Schools. (2025). Pandas DataFrame isin() Method. https://www.w3schools.com/python/pandas/ref_df_isin.asp

7. **drop() method:**
   W3Schools. (2025). Pandas DataFrame drop() Method.
   https://www.w3schools.com/python/pandas/ref_df_drop.asp

8. **Create a line-bar chart with double axis:**

   - Name, version, company of AI tool: Chat GPT-5 mini.

   - Chat objective: to learn how to create a line-bar chart with 2 axis

   - Use of AI-generated content:

   Evaluation of Results: ChatGPT correctly taught me how to create multiple-axis subplots, customize the primary and secondary axis (e.g., font size, rotation degree), and wrap text on x-axis.

   Limitation: Its first suggestion on how to apply the text wrapping *textwrap.wrap()* did not live up to my expectation as the output text was still quite long and we needed to manually add a line breaker (\n). I then asked it for other ways and ChatGPT returned with quite useful ideas e.g., changing the font size or using another function called *textwrap.fill()* which breaks lines directly.

   How I used the AI's response: I used the AI's guidance as a reference point to create a customized chart with my own data.

   - URL of chat history:
   https://chatgpt.com/share/68e01fdf-c494-8009-a7a1-fb4f53788a67

   - Time and date of chat: Oct 3 12pm

LITERATURE CITATION

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